

## Temperature Measurement Uni Curve Miniature Sensors

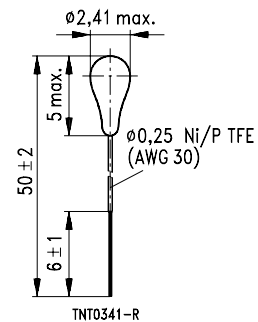
**B57863**  
**S 863**

### Applications

- Heating and air conditioning systems
- Industrial electronics
- Automotive electronics

### Features

- Fast response
- Accuracy of 0,2 K between 0 °C and 70 °C
- Excellent long-term stability
- Uni curve sensor
- Epoxy resin encapsulation
- PTFE-insulated leads of nickel wire, AWG 30



Dimensions in mm  
Approx. weight 60 mg

|                                        |               |                    |      |
|----------------------------------------|---------------|--------------------|------|
| Climatic category (IEC 68-1)           |               | 55/155/56          |      |
| Max. power at 25 °C                    | $P_{25}$      | 60                 | mW   |
| Temperature tolerance (0 ... 70 °C)    | $\Delta T$    | $\pm 0,2, \pm 0,5$ | K    |
| Rated temperature                      | $T_N$         | 25                 | °C   |
| Dissipation factor (in air)            | $\delta_{th}$ | approx. 1,5        | mW/K |
| Thermal cooling time constant (in air) | $\tau_c$      | approx. 15         | s    |
| Heat capacity                          | $C_{th}$      | approx. 22,5       | mJ/K |

| Type            | $R_{25}$<br>$\Omega$ | No. of $R/T$<br>characteristic | $B_{25/100}$<br>K | Ordering code   |
|-----------------|----------------------|--------------------------------|-------------------|-----------------|
| S 863/3 k/+ 40  | 3 k                  | 8016                           | 3988              | B57863-S302-+40 |
| S 863/5 k/+ 40  | 5 k                  | 8016                           | 3988              | B57863-S502-+40 |
| S 863/10 k/+ 40 | 10 k                 | 8016                           | 3988              | B57863-S103-+40 |
| S 863/30 k/+ 40 | 30 k                 | 8018                           | 3964              | B57863-S303-+40 |

+: F for  $\Delta T = 0,2$  K  
G for  $\Delta T = 0,5$  K

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**R/T characteristics**

| Number        | <b>8016</b>                  | <b>8016</b>                  | <b>8016</b>                   | <b>8018</b>                   |
|---------------|------------------------------|------------------------------|-------------------------------|-------------------------------|
| <i>T</i> (°C) | $R_{25} = 3 \text{ k}\Omega$ | $R_{25} = 5 \text{ k}\Omega$ | $R_{25} = 10 \text{ k}\Omega$ | $R_{25} = 30 \text{ k}\Omega$ |
|               | $R_T (\Omega)$               | $R_T (\Omega)$               | $R_T (\Omega)$                | $R_T (\Omega)$                |
| – 55,0        | 288910                       | 481520                       | 963000                        | –                             |
| – 50,0        | 201030                       | 335050                       | 670100                        | –                             |
| – 45,0        | 141510                       | 235840                       | 471700                        | –                             |
| – 40,0        | 100950                       | 168250                       | 336500                        | 907200                        |
| – 35,0        | 72777                        | 121290                       | 242600                        | 663000                        |
| – 30,0        | 53100                        | 88500                        | 177000                        | 489600                        |
| – 25,0        | 39111                        | 65185                        | 130400                        | 365100                        |
| – 20,0        | 29121                        | 48535                        | 97070                         | 274590                        |
| – 15,0        | 21879                        | 36465                        | 72930                         | 208350                        |
| – 10,0        | 16599                        | 27665                        | 55330                         | 159390                        |
| – 5,0         | 12695                        | 21158                        | 42320                         | 122910                        |
| 0,0           | 9795,0                       | 16325                        | 32650                         | 954490                        |
| 1,0           | 9308,1                       | 15514                        | 31030                         | 90870                         |
| 2,0           | 8848,5                       | 14747                        | 29490                         | 86490                         |
| 3,0           | 8414,3                       | 14024                        | 28050                         | 82350                         |
| 4,0           | 8004,0                       | 13340                        | 26680                         | 78420                         |
| 5,0           | 7616,3                       | 12694                        | 25390                         | 74730                         |
| 6,0           | 7249,6                       | 12083                        | 24170                         | 71220                         |
| 7,0           | 6902,8                       | 11505                        | 23010                         | 67890                         |
| 8,0           | 6574,7                       | 10958                        | 21920                         | 64710                         |
| 9,0           | 6264,1                       | 10440                        | 20880                         | 61740                         |
| 10,0          | 5970,0                       | 9950,0                       | 19900                         | 58890                         |
| 11,0          | 5690,7                       | 9484,5                       | 18970                         | 56190                         |
| 12,0          | 5426,1                       | 9043,5                       | 18090                         | 53640                         |
| 13,0          | 5175,4                       | 8625,6                       | 17250                         | 51210                         |
| 14,0          | 4937,7                       | 8229,5                       | 16460                         | 48900                         |
| 15,0          | 4712,3                       | 7853,9                       | 15710                         | 46710                         |
| 16,0          | 4498,6                       | 7497,6                       | 15000                         | 44640                         |
| 17,0          | 4295,8                       | 7159,6                       | 14320                         | 42660                         |
| 18,0          | 4103,3                       | 6383,8                       | 13680                         | 40800                         |
| 19,0          | 3920,6                       | 6534,2                       | 13070                         | 39030                         |
| 20,0          | 3747,0                       | 6245,0                       | 12490                         | 37320                         |
| 21,0          | 3582,1                       | 5970,1                       | 11940                         | 35700                         |
| 22,0          | 3425,3                       | 5708,9                       | 11420                         | 34170                         |
| 23,0          | 3276,4                       | 5460,6                       | 10920                         | 32730                         |
| 24,0          | 3134,7                       | 5224,5                       | 10450                         | 31320                         |
| 25,0          | 3000,0                       | 5000,0                       | 10000                         | 30000                         |

**R/T characteristics**

| Number        | <b>8016</b>                  | <b>8016</b>                  | <b>8016</b>                   | <b>8018</b>                   |
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| <i>T</i> (°C) | $R_{25} = 3 \text{ k}\Omega$ | $R_{25} = 5 \text{ k}\Omega$ | $R_{25} = 10 \text{ k}\Omega$ | $R_{25} = 30 \text{ k}\Omega$ |
|               | $R_T (\Omega)$               | $R_T (\Omega)$               | $R_T (\Omega)$                | $R_T (\Omega)$                |
| 26,0          | 2871,6                       | 4786,0                       | 9572                          | 28737                         |
| 27,0          | 2749,5                       | 4582,5                       | 9165                          | 27531                         |
| 28,0          | 2633,2                       | 4388,7                       | 8777                          | 26385                         |
| 29,0          | 2522,5                       | 4204,2                       | 8408                          | 25290                         |
| 30,0          | 2417,1                       | 4028,5                       | 8057                          | 24249                         |
| 31,0          | 2316,5                       | 3860,9                       | 7722                          | 23256                         |
| 32,0          | 2220,7                       | 3701,2                       | 7402                          | 22305                         |
| 33,0          | 2129,4                       | 3549,1                       | 7098                          | 21402                         |
| 34,0          | 2042,4                       | 3404,0                       | 6808                          | 20538                         |
| 35,0          | 1959,4                       | 3265,7                       | 6531                          | 19716                         |
| 36,0          | 1880,2                       | 3133,7                       | 6267                          | 18927                         |
| 37,0          | 1804,7                       | 3007,8                       | 6016                          | 18177                         |
| 38,0          | 1732,6                       | 2887,7                       | 5775                          | 17460                         |
| 39,0          | 1663,8                       | 2773,0                       | 5546                          | 16773                         |
| 40,0          | 1598,1                       | 2663,5                       | 5327                          | 16119                         |
| 41,0          | 1535,2                       | 2558,7                       | 5117                          | 15495                         |
| 42,0          | 1475,2                       | 2458,7                       | 4917                          | 14895                         |
| 43,0          | 1417,8                       | 2363,1                       | 4726                          | 14325                         |
| 44,0          | 1363,0                       | 2271,7                       | 4543                          | 13776                         |
| 45,0          | 1310,6                       | 2184,4                       | 4369                          | 13254                         |
| 46,0          | 1260,5                       | 2100,8                       | 4202                          | 12750                         |
| 47,0          | 1212,6                       | 2021,0                       | 4042                          | 12273                         |
| 48,0          | 1166,7                       | 1944,6                       | 3889                          | 11811                         |
| 49,0          | 1122,9                       | 1871,5                       | 3743                          | 11373                         |
| 50,0          | 1080,9                       | 1801,5                       | 3603                          | 10950                         |
| 51,0          | 1040,6                       | 1734,4                       | 3469                          | 10545                         |
| 52,0          | 1002,0                       | 1670,1                       | 3340                          | 10158                         |
| 53,0          | 965,12                       | 1608,5                       | 3217                          | 9786,0                        |
| 54,0          | 929,75                       | 1549,6                       | 3099                          | 9429,0                        |
| 55,0          | 895,86                       | 1493,1                       | 2986                          | 9090,0                        |
| 56,0          | 863,40                       | 1439,0                       | 2878                          | 8760,0                        |
| 57,0          | 832,28                       | 1387,1                       | 2774                          | 8448,0                        |
| 58,0          | 802,44                       | 1337,4                       | 2675                          | 8148,0                        |
| 59,0          | 773,84                       | 1289,7                       | 2579                          | 7857,0                        |
| 60,0          | 746,40                       | 1244,0                       | 2488                          | 7581,0                        |
| 61,0          | 720,07                       | 1200,1                       | 2400                          | 7314,0                        |
| 62,0          | 694,81                       | 1158,0                       | 2316                          | 7059,0                        |
| 63,0          | 670,56                       | 1117,6                       | 2235                          | 6818,0                        |
| 64,0          | 647,27                       | 1078,8                       | 2158                          | 6579,0                        |
| 65,0          | 624,91                       | 1041,5                       | 2083                          | 6354,0                        |

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| <i>T</i> (°C) | $R_{25} = 3 \text{ k}\Omega$ | $R_{25} = 5 \text{ k}\Omega$ | $R_{25} = 10 \text{ k}\Omega$ | $R_{25} = 30 \text{ k}\Omega$ |
|               | $R_T (\Omega)$               | $R_T (\Omega)$               | $R_T (\Omega)$                | $R_T (\Omega)$                |
| 66,0          | 603,43                       | 1005,7                       | 2011                          | 6135,0                        |
| 67,0          | 582,80                       | 971,33                       | 1943                          | 5928,0                        |
| 68,0          | 562,97                       | 938,29                       | 1877                          | 5727,0                        |
| 69,0          | 543,92                       | 906,53                       | 1813                          | 5535,0                        |
| 70,0          | 525,60                       | 876,00                       | 1752                          | 5349,0                        |
| 75,0          | 444,41                       | 740,69                       | 1481                          | 4524,0                        |
| 80,0          | 377,40                       | 629,00                       | 1258                          | 3840,0                        |
| 85,0          | 321,70                       | 536,17                       | 1072                          | 3273,0                        |
| 90,0          | 275,31                       | 458,85                       | 917,7                         | 2799,0                        |
| 95,0          | 236,56                       | 394,26                       | 788,5                         | 2404,8                        |
| 100,0         | 204,00                       | 340,00                       | 680,0                         | 2073,0                        |
| 105,0         | 176,58                       | 294,30                       | 588,6                         | 1792,2                        |
| 110,0         | 153,36                       | 255,60                       | 511,2                         | 1554,9                        |
| 115,0         | 133,62                       | 222,70                       | 445,4                         | 1353,6                        |
| 120,0         | 116,79                       | 194,65                       | 389,3                         | 1182,0                        |
| 125,0         | 102,51                       | 170,85                       | 341,7                         | 1035,0                        |
| 130,0         | 90,270                       | 150,45                       | 300,9                         | 909,60                        |
| 135,0         | 79,633                       | 132,72                       | 265,4                         | 801,60                        |
| 140,0         | 70,440                       | 117,40                       | 234,8                         | 708,30                        |
| 145,0         | 62,496                       | 104,16                       | 208,3                         | 627,30                        |
| 150,0         | 55,590                       | 92,650                       | 185,3                         | 557,10                        |
| 155,0         | 49,604                       | 82,674                       | 165,3                         | –                             |